Signed accuracy contrasts over acuity and colony size

Studies 1–7 fix specific parameter configurations (sensor acuity and colony size) to isolate mechanistic claims.

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Signed accuracy contrasts... (part 2)

Study 8 asks how the communicating-minus-isolated and hierarchical-minus-flat accuracy contrasts vary across a systematic two-dimensional sweep.

Signed accuracy contrasts... (part 3)

This estimand-first formulation makes sign changes visible without presuming that either architecture is uniformly advantageous.

We sweep sensor acuity $\kappa \in \{0.40, 0.55, 0.70, 0.85, 0.95\}$ and colony size $n \in \{2, 4, 6, 8, 10\}$, evaluating two systems:

Signed accuracy contrasts... (part 4)

- **Belief sharing** (Study 1 architecture) — accuracy gap = communicating minus isolated mean accuracy;
- **Hierarchical POMDP** (Study 6 architecture) — accuracy gap = hierarchical minus flat location accuracy.

Signed accuracy contrasts... (part 5)

Each cell averages $n_{\text{trials}} = 20$ independent trials to reduce Monte-Carlo noise at the cell level.

Signed accuracy contrasts… (part 6)

The resulting heatmaps (fig. 32) show the accuracy gap for both systems as a function of acuity and colony size. The signed scale is centered on zero: brown cells encode positive gaps, blue cells encode negative gaps, and the neutral midpoint encodes zero.

Signed accuracy contrasts... (part 7)

Every cell also prints its signed value, so the direction remains explicit without color. Hatching identifies only the declared $|\mathrm{gap}| \leq 0.05$ display band; it is not an unreliability flag, confidence interval, significance test, or proof of zero effect.

Signed accuracy contrasts... (part 8)

Across the grid the following patterns hold:

Pattern 1: low-to-moderate acuity

Positive communicating-minus-isolated contrasts occur at low-to-moderate acuity. The accuracy gap peaks in the second-lowest acuity row, where observations carry some signal but no single sentinel resolves the location alone.

Pattern 1: low-to-moderate... (part 2)

It remains positive across the two lowest-acuity rows for colonies of at least four agents in the declared grid.

Near ceiling acuity the gap approaches zero because isolated agents already solve most trials in the declared task.

Pattern 2: a colony-size floor

Colony size acts through a floor, not a smooth slope. The two-agent column shows an exactly zero belief-sharing gap at every acuity. Under self-exclusion ("agents do not hear themselves"), each member hears exactly one incoming belief, so the heard consensus adds no pooled evidence.

Pattern 2: a colony-size... (part 2)

Colonies of four or more show the positive low-acuity contrast within this grid.

Pattern 3: a conditional hierarchical contrast

The hierarchical contrast changes with acuity and colony size. Negative hierarchical-minus-flat location-accuracy gaps occur across the lower-acuity rows, including colonies larger than two agents.

Pattern 3: a conditional... (part 2)

The higher-acuity rows are near zero outside the two-agent column; positive cells are confined to that column in this grid. This conditional pattern does not establish a general hierarchical accuracy advantage,

Pattern 3: a conditional... (part 3)

and the per-study paired location contrast remains a separate estimate under its own trial and seed budget.

Pattern 3: a conditional... (part 4)

The full parameter grid and protocol details are in the supplement (sec. 17.14). A native-unit cross-study overview of the headline metrics across all 9 studies is shown in fig. 33.

Pattern 3: a conditional... (part 5)

Read the page-compatible composition down the full left column and then down the right: Panel A contains the six accuracy rows, Panel B at upper right the two information or free-energy rows, and Panel C at lower right the single recovery row. Each study appears once, in its applicable native-unit panel.

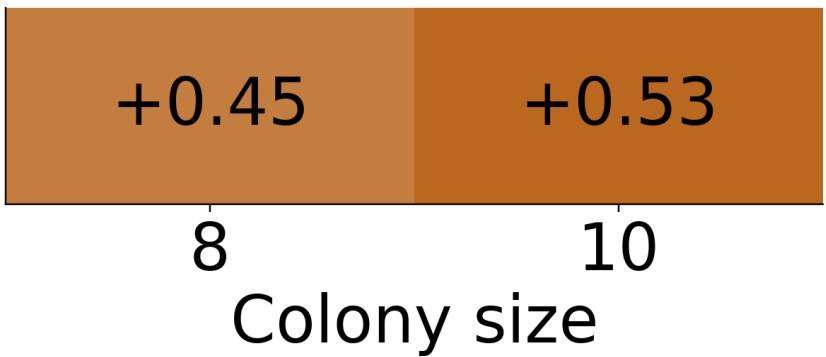
Pattern 3: a conditional... (part 6)

Sharing: acuity 0.40



Pattern 3: a conditional... (part 7)

Sharing: acuity 0.40



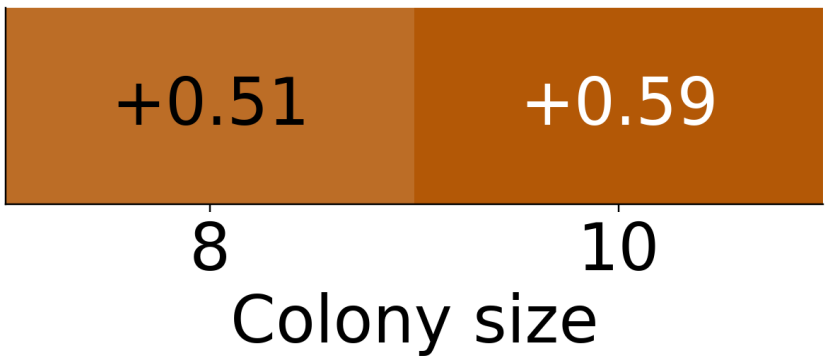
Pattern 3: a conditional... (part 8)

Sharing: acuity 0.55



Pattern 3: a conditional... (part 9)

Sharing: acuity 0.55



Pattern 3: a conditional... (part 10)

Sharing: acuity 0.70



Pattern 3: a conditional… (part 11)

Sharing: acuity 0.70



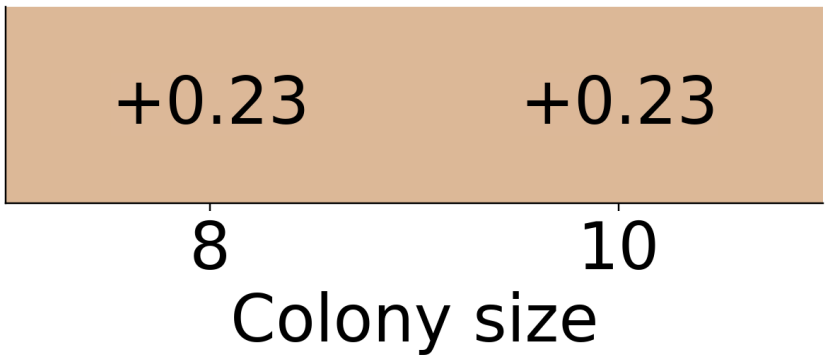
Pattern 3: a conditional… (part 12)

Sharing: acuity 0.85



Pattern 3: a conditional... (part 13)

Sharing: acuity 0.85



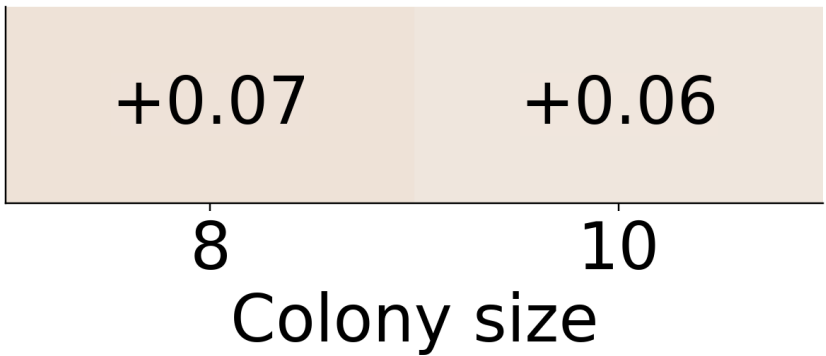
Pattern 3: a conditional… (part 14)

Sharing: acuity 0.95



Pattern 3: a conditional… (part 15)

Sharing: acuity 0.95

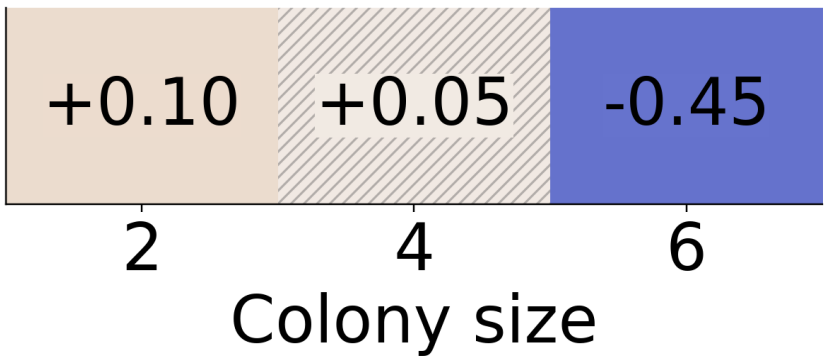


Pattern 3: a conditional... (part 16)

Sharing: max |gap| = 0.586; positive cells above the display band = 19/25.

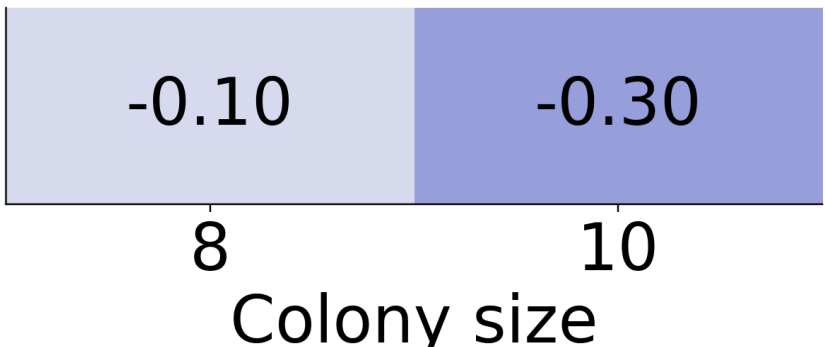
Pattern 3: a conditional... (part 17)

Hierarchy: acuity 0.40



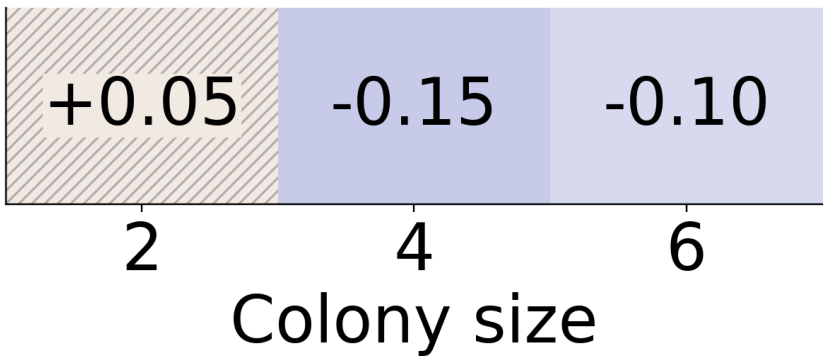
Pattern 3: a conditional... (part 18)

Hierarchy: acuity 0.40



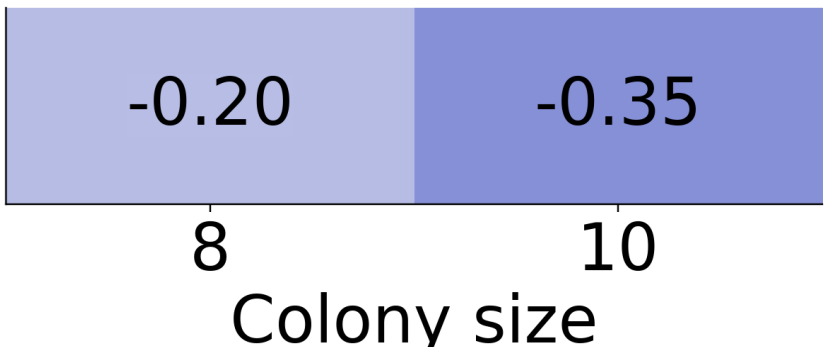
Pattern 3: a conditional... (part 19)

Hierarchy: acuity 0.55



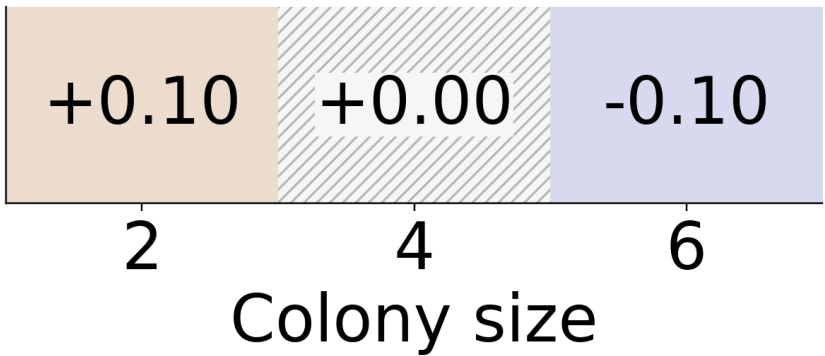
Pattern 3: a conditional... (part 20)

Hierarchy: acuity 0.55



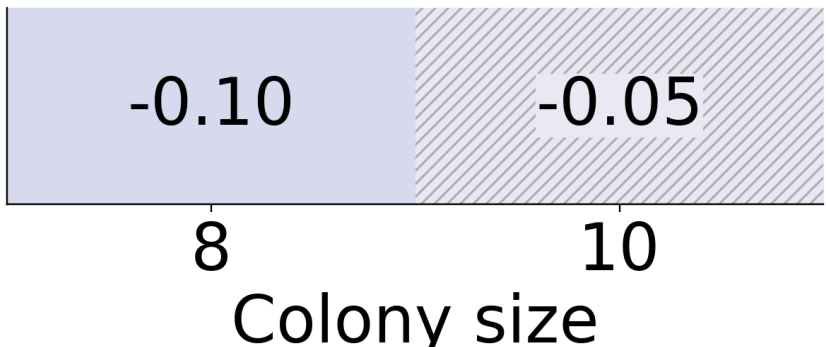
Pattern 3: a conditional... (part 21)

Hierarchy: acuity 0.70



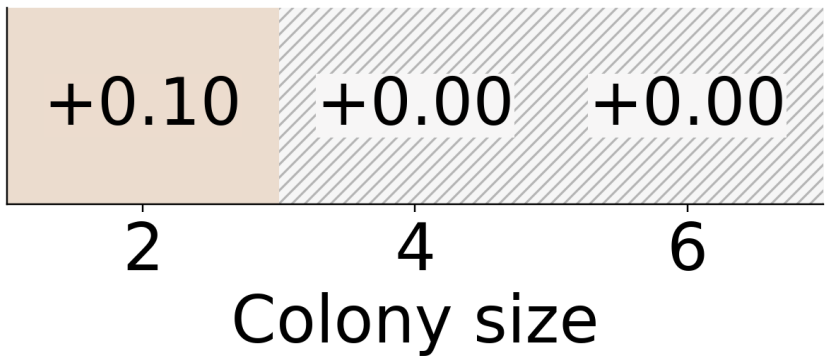
Pattern 3: a conditional... (part 22)

Hierarchy: acuity 0.70



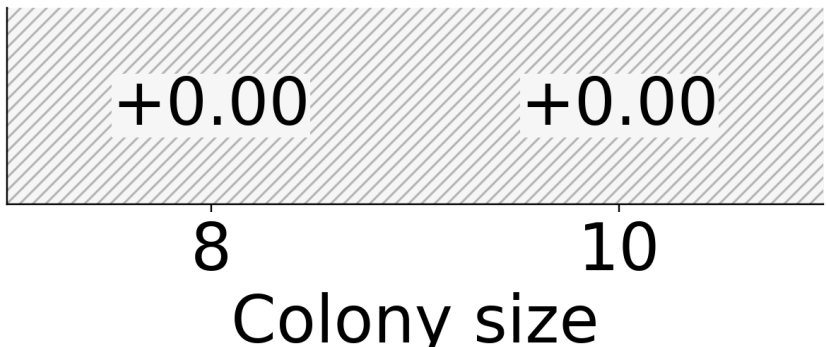
Pattern 3: a conditional... (part 23)

Hierarchy: acuity 0.85



Pattern 3: a conditional... (part 24)

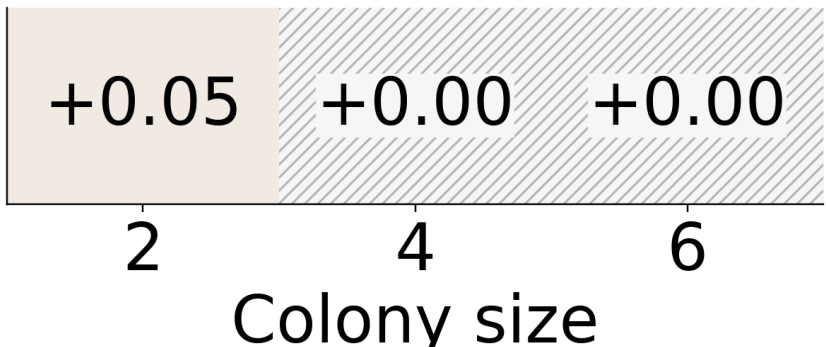
Hierarchy: acuity 0.85



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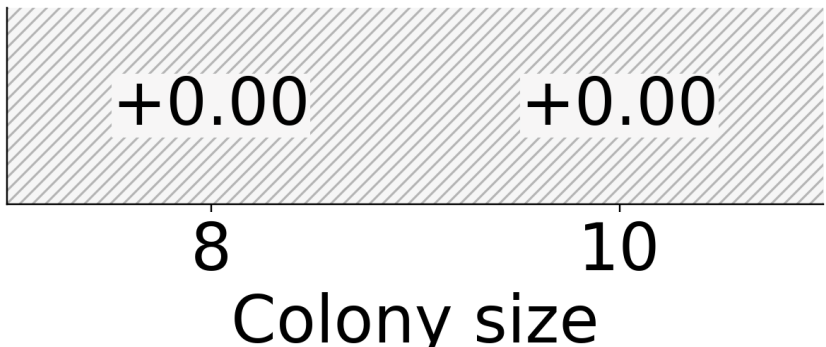
Pattern 3: a conditional... (part 25)

Hierarchy: acuity 0.95



Pattern 3: a conditional... (part 26)

Hierarchy: acuity 0.95



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Pattern 3: a conditional... (part 27)

Hierarchy: max |gap| = 0.450; positive cells above the display band = 4/25.

Pattern 3: a conditional... (part 28)

Sharing: communicating minus isolated accuracy. Hierarchy: hierarchical minus flat location accuracy. Positive and negative values retain their sign.

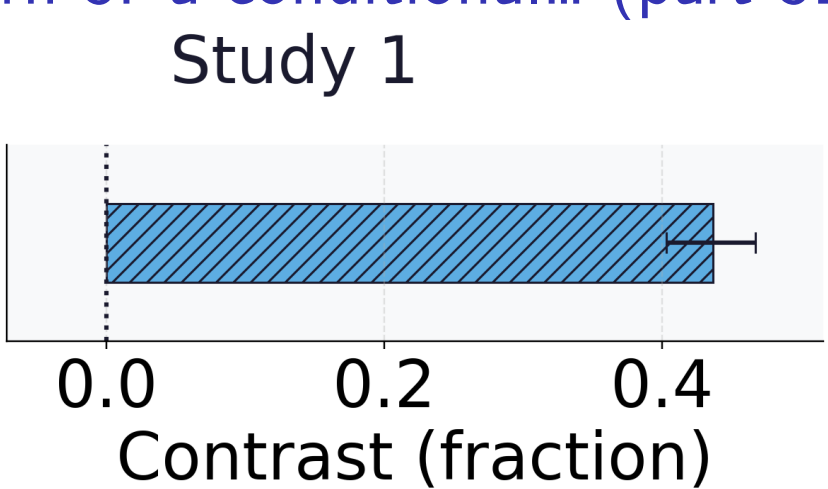
Pattern 3: a conditional... (part 29)

Common signed color scale: [-0.5861, 0.5861]. Every cell prints its value; panels keep original acuity and colony ordering.

Pattern 3: a conditional… (part 30)

Hatching uses unrounded gaps: |gap| ≤ 0.05. Printed values are rounded; equal printed values can have different hatching. It is not a CI, significance test, unreliability flag or proof of zero effect.

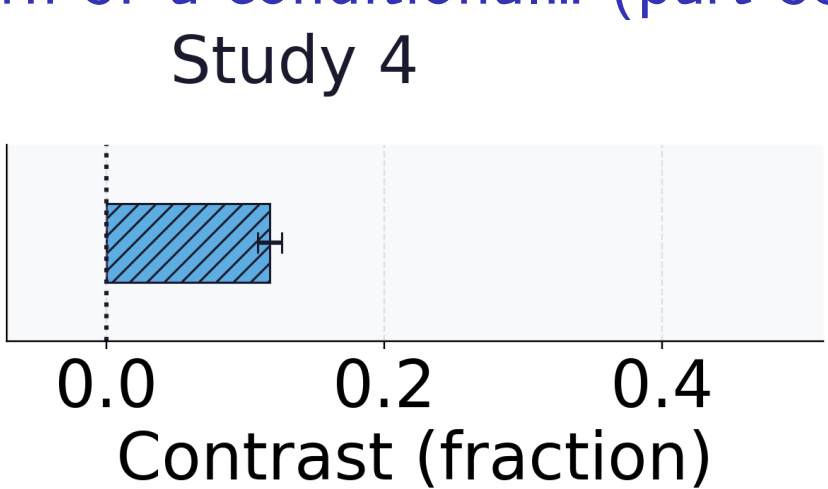
Pattern 3: a conditional... (part 31)



Pattern 3: a conditional... (part 32)

Study 1 Belief sharing. Mean +0.437; 95% interval [+0.4034, +0.4675] fraction.

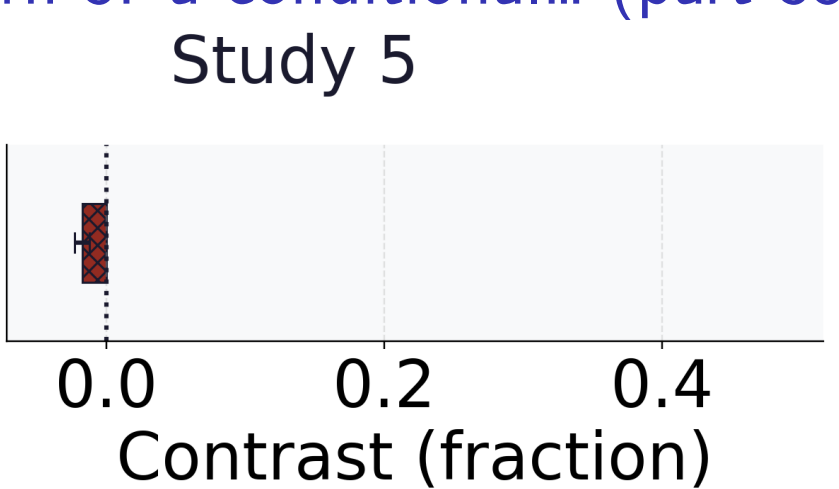
Pattern 3: a conditional... (part 33)



Pattern 3: a conditional… (part 34)

Study 4 Within-run display-selected maximum. Mean +0.1175; 95% interval [+0.109, +0.1262] fraction.

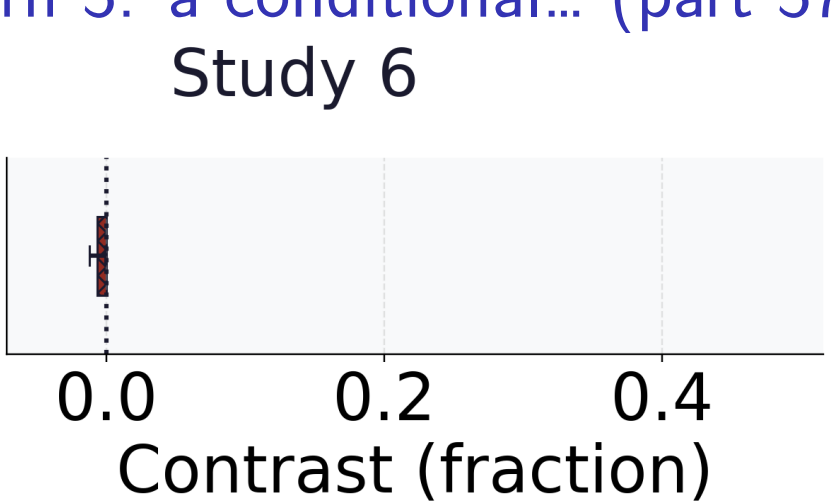
Pattern 3: a conditional... (part 35)



Pattern 3: a conditional... (part 36)

Study 5 Moving world (EFE). Mean -0.01758; 95% interval [-0.02305, -0.0125] fraction.

Pattern 3: a conditional... (part 37)



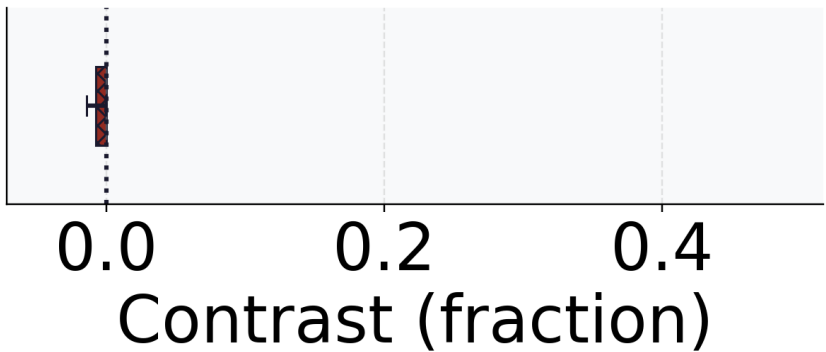
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Pattern 3: a conditional... (part 38)

Study 6 2-level hierarchical POMDP. Mean -0.006641; 95% interval [-0.0125, -0.001172] fraction.

Pattern 3: a conditional... (part 39)

Study 7

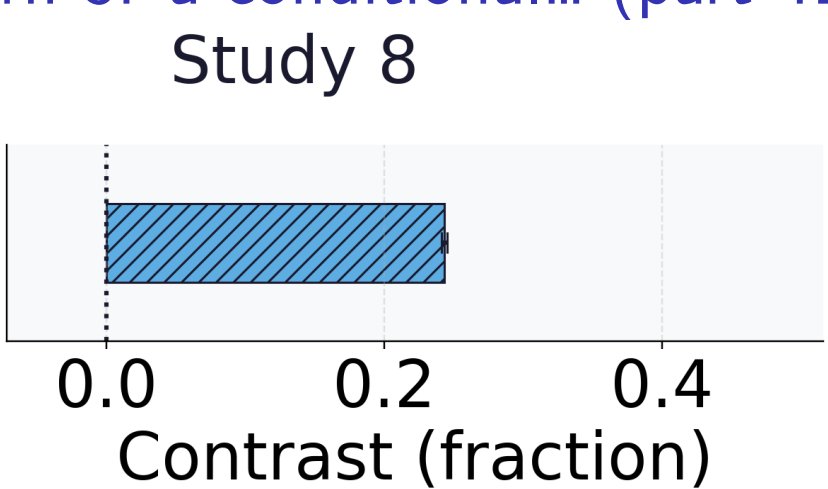


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Pattern 3: a conditional... (part 40)

Study 7 3-level hierarchical POMDP. Mean -0.007812; 95% interval [-0.01445, -0.0007813] fraction.

Pattern 3: a conditional... (part 41)

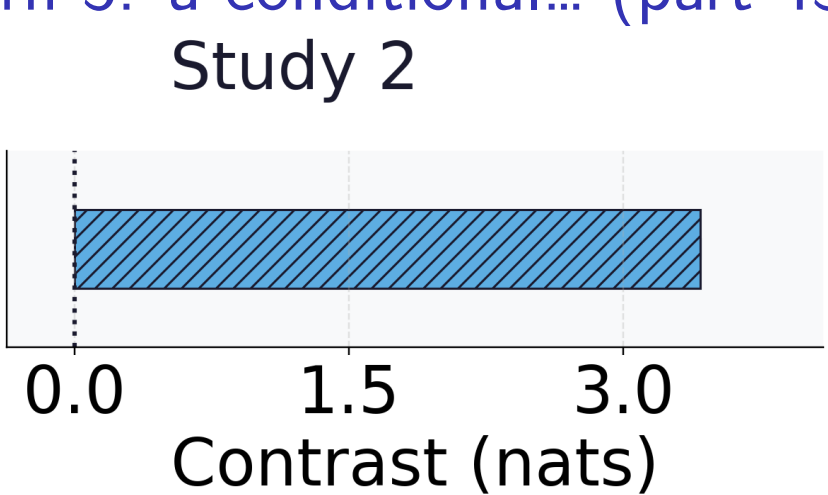


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Pattern 3: a conditional... (part 42)

Study 8 Sensitivity sweep. Mean +0.2433; 95% interval [+0.2411, +0.2455] fraction.

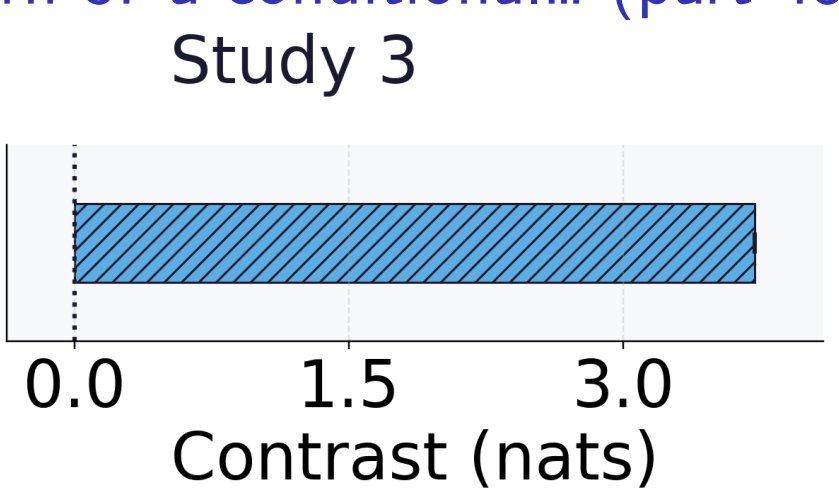
Pattern 3: a conditional... (part 43)



Pattern 3: a conditional... (part 44)

Study 2 Language acquisition. Mean +3.42; 95% interval [+3.42, +3.421] nats.

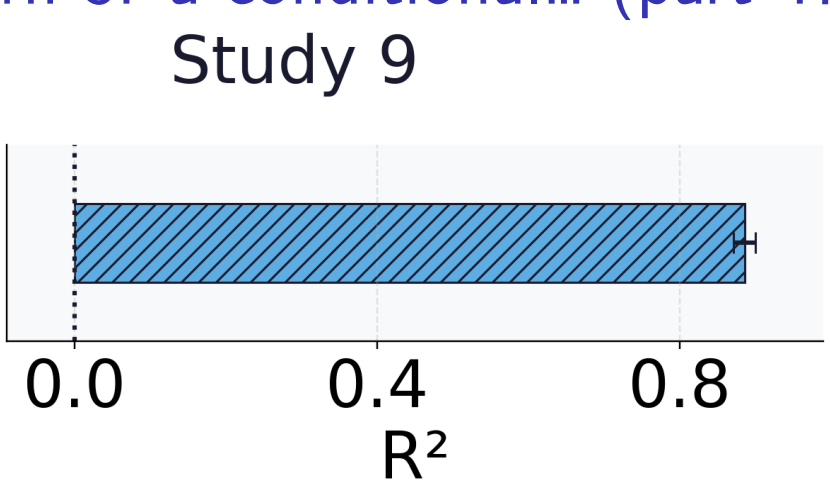
Pattern 3: a conditional... (part 45)



Pattern 3: a conditional... (part 46)

Study 3 Configured BMR sign control. Mean +3.716; 95% interval [+3.71, +3.722] nats.

Pattern 3: a conditional... (part 47)



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Pattern 3: a conditional... (part 48)

Study 9 Parameter recovery. Mean +0.8861; 95% interval [+0.8716, +0.9005] R-sq.

Pattern 3: a conditional… (part 49)

Separate harmonized seed-level rerun: n = 128 independent seeds. Whiskers are 95% percentile-bootstrap intervals.

Pattern 3: a conditional… (part 50)

/// positive; xxx negative; dots within 0.001 of zero. Hatches describe the mean sign, not statistical significance. Direction remains metric-specific.

Pattern 3: a conditional… (part 51)

Native-unit facets stay separate. R^2 is a fit statistic, not an effect contrast. No cross-unit ranking; Study 4 is a within-run display-selected maximum.